SwiftLog Executive Summary

Machine Learning Track | Week 5

Objective

Build an educational machine-learning prototype to explore delivery delays, product returns, weekly product demand, and next-month customer revenue using the supplied SwiftLog datasets.

What was completed

Reviewed and cleaned source tables; engineered task-specific features; compared three algorithms for each of four tasks; evaluated held-out predictions; saved selected model pipelines; generated comparison tables and evaluation visuals; and prepared a Streamlit demonstration app.

Results at a glance

Problem	Selected algorithm	F1	ROC-AUC	MAE	RMSE	R2
Delivery Delay Classification	Random Forest (tuned)	0.904				
Product Return Classification	Decision Tree	0.161				
Product Demand Forecasting	Random Forest Regressor (tuned)			6.404	7.957	-0.028
Customer Value Prediction (Next Month)	Linear Regression			3160.402	4082.078	0.001

What the results mean

The current models are not ready for operational use. The delay classifier has high F1 but ROC-AUC close to 0.5, suggesting weak class separation. The return classifier also shows weak discrimination. Demand and next-month customer-value regression have negative held-out R-squared, so they do not outperform a simple mean baseline on the test period.

Recommended next steps

Collect richer historical data, verify target definitions, add operationally available predictors, compare against simple baselines, and validate on later time periods. Use the app only as a demonstration and keep human review in the decision process.

Conclusion

The project delivers a complete prototype workflow and a documented evaluation. The most important outcome is identifying the data and validation improvements needed before any business deployment.